The Factor Graph Approach to Model-Based Estimation in Robotics

Hugo Aguettaz

October 29, 2026

Chapter 1

Introduction

Throughout this report, we will heavily use factor graphs, both for reasoning and for describing algorithms. We will use factor graphs as in [**?**, **?**], where nodes/boxes represent factors and edges represent variables. Of particular interest are the tabulated message-passing rules therein.

Chapter 2

MAP Estimation

By forward-filtering, backward-deciding.

Chapter 3

Marginalization

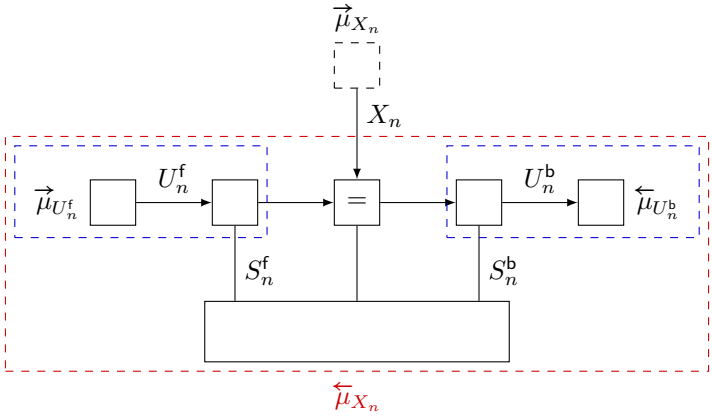


Figure 3.1: Marginalization.

We consider computing the marginal distributions in a cyclic factor graph.

As illustrated in Figure 3.3, the marginal distribution of X_n can be decomposed as:

$$p_{X_n}(x_n) \propto \vec{\mu}_{X_n}(x_n)\, \overleftarrow{\mu}_{X_n}(x_n) \tag{3.1}$$

where $\vec{\mu}_{X_n}(x_n)$ are sum-product messages $\overleftarrow{\mu}_{X_n}(x_n)$, with $\vec{\mu}_{X_n}$ summarizing all local factors depending solely on X_n while $\overleftarrow{\mu}_{X_n}$ summarizes the contributions of all remaining factors.

To compute $\overleftarrow{\mu}_{X_n}(x_n)$ three mutually independent components are necessary:

1. A forward message obtained by marginalizing out U_n^{f}, where U_n^{f} are forward variables (i.e., X_ℓ for $\ell < n$) isolated from the backward subsystem, and S_n^{f} are those coupled to it.

2. A backward message obtained by marginalizing out U_n^{b}, where U_n^{b} are backward variables (i.e., X_ℓ for $\ell > n$) isolated from the forward subsystem, and S_n^{b} are those coupled to it.

3. A coupling factor encompassing all factors bridging S_n^{f} and S_n^{b} (with and without X_n).

In practice, the forward and backward messages are computed recursively via separate forward and backward passes.

3.1 The Gaussian Case

3.1.1 Gaussian Factor



Figure 3.2: A Gaussian prior (left) and a Gaussian observation model (right).

In this report, we focus exclusively on single-variable Gaussian priors and linear Gaussian observation models, both illustrated in Figure 3.2. The observation model represents a linear Gaussian observation of X via AX.

We parameterize these models in canonical information form, using precision matrices and information vectors. Standard parameters (mean vectors and covariance matrices) are only used during the initial construction of the factor graph or at the final stage to recover marginal distributions.

3.1.2 Marginal Distribution

The marginal distribution of X, given a prior and an observation model (as depicted in Figure 3.3), is computed using standard Gaussian message passing rules [**?**]:

$$\xi_X = \overrightarrow{\xi}_X + A^{\mathsf{T}} \overleftarrow{\xi}_Y \tag{3.2}$$

$$W_X = \overrightarrow{W}_X + A^{\mathsf{T}} \overleftarrow{W}_Y A. \tag{3.3}$$

If the posterior precision matrix W_X is invertible (i.e., strictly positive definite), the posterior mean m_X is uniquely determined by solving the linear system $W_X m_X = \xi_X$. For this step, a standard Cholesky decomposition is the most computationally efficient approach, and the resulting factorization can be directly reused to recover the posterior covariance matrix V_X $(= W_X^{-1})$.

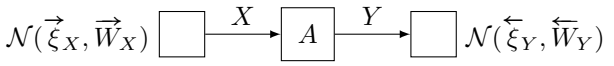


Figure 3.3: Factor graph representing the marginalization of X given a prior and a linear observation model.

3.1.3 Augmented Observation Models

Consider two observation models, $Y_1 = A_1X_1 + B_1U$ and $Y_2 = A_2X_2 + B_2U$, coupled via a common variable U. These observations can be combined into a single augmented observation model:

$$\underbrace{\begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix}}_{Y} = \underbrace{\begin{bmatrix} A_1 & 0 & B_1 \\ 0 & A_2 & B_2 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} X_1 \\ X_2 \\ U \end{bmatrix}}_{X} \tag{3.4}$$

with Gaussian parameters defined as:

$$\overleftarrow{\xi}_Y \triangleq \begin{bmatrix} \overleftarrow{\xi}_{Y_1} \\ \overleftarrow{\xi}_{Y_2} \end{bmatrix} \quad \text{and} \quad \overleftarrow{W}_Y \triangleq \begin{bmatrix} \overleftarrow{W}_{Y_1} & 0 \\ 0 & \overleftarrow{W}_{Y_2} \end{bmatrix} \tag{3.5}$$

This formulation naturally generalizes to an arbitrary number of observation models. In practice, neither A nor $\overleftarrow{W}_Y$ should be stored as dense matrices. Exploiting their inherent sparsity is essential to minimize both memory footprint and computational overhead.

3.1.4 Partial Box Closure

Consider the partial closure of an observation model $Y = AX + BU$ with respect to the variable U. We assume U has a prior model independent of X, parameterized by an information vector $\vec{\xi}_U$ and a precision matrix $\vec{W}_U$.

Marginalizing out U and Y (closing the blue dashed box in Figure 3.4) yields a new Gaussian observation model that depends exclusively on X. Applying the standard Gaussian message-passing rules [**?**], its parameters are

$$\overleftarrow{\xi}_Z = \overleftarrow{\xi}_Y - \overleftarrow{W}_Y B H \big(\vec{\xi}_U + B^\mathsf{T} \overleftarrow{\xi}_Y\big), \tag{3.6}$$

$$\overleftarrow{W}_Z = \overleftarrow{W}_Y - \overleftarrow{W}_Y B H B^\mathsf{T} \overleftarrow{W}_Y, \tag{3.7}$$

with

$$H \triangleq \big(\vec{W}_U + B^\mathsf{T} \overleftarrow{W}_Y B\big)^{-1}. \tag{3.8}$$

Numerical implementation. H should never be formed explicitly. Instead, let LL^T be the Cholesky factorization of $\vec{W}_U + B^\mathsf{T} \overleftarrow{W}_Y B$, and define

$$G \triangleq L^{-1} B^\mathsf{T} \overleftarrow{W}_Y \tag{3.9}$$

$$h \triangleq L^{-1} \big(\vec{\xi}_U + B^\mathsf{T} \overleftarrow{\xi}_Y\big), \tag{3.10}$$

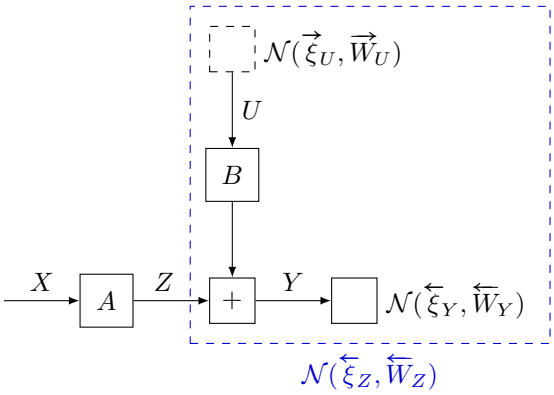


Figure 3.4: Factor graph ....

which require only two triangular solves. Since $H = L^{-\mathsf{T}}L^{-1}$ and $\overleftarrow{W}_Y$ is symmetric, we obtain

$$\overleftarrow{\xi}_Z = \overleftarrow{\xi}_Y - G^{\mathsf{T}}h, \tag{3.11}$$

$$\overleftarrow{W}_Z = \overleftarrow{W}_Y - G^{\mathsf{T}}G. \tag{3.12}$$

If the matrix is not positive definite, Tikhonov regularization can be applied by replacing $\vec{W}_U$ with $\vec{W}_U + \epsilon I$ for a small $\epsilon > 0$. This is equivalent to adding a weak prior $\mathcal{N}(0, \epsilon^{-1}I)$ on U.

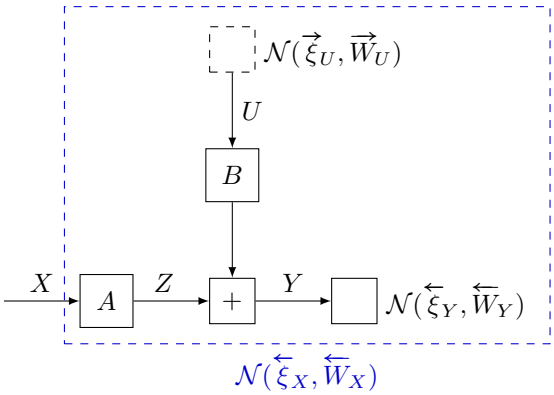


Figure 3.5: Factor graph ....

When $\dim Z > \dim X$, it is recommended to close the box including A, as shown in Figure 3.5. As before, let LL^{T} be the Cholesky factorization of

$\overrightarrow{W}_U + B^\mathsf{T}\overleftarrow{W}_Y B$, and define

$$F \triangleq L^{-1}B^\mathsf{T}\overleftarrow{W}_Y A \tag{3.13}$$

$$h \triangleq L^{-1}\big(\overrightarrow{\xi}_U + B^\mathsf{T}\overleftarrow{\xi}_Y\big), \tag{3.14}$$

which require only two triangular solves. Since $H = L^{-\mathsf{T}}L^{-1}$ and $\overleftarrow{W}_Y$ is symmetric, we obtain

$$\overleftarrow{\xi}_X = A^\mathsf{T}\overleftarrow{\xi}_Z \tag{3.15}$$

$$= A^\mathsf{T}\overleftarrow{\xi}_Y - F^\mathsf{T}h, \tag{3.16}$$

$$\overleftarrow{W}_X = A^\mathsf{T}\overleftarrow{W}_Z A \tag{3.17}$$

$$= A^\mathsf{T}\overleftarrow{W}_Y A - F^\mathsf{T}F. \tag{3.18}$$

3.1.5 Practical Considerations

We conclude this section with practical advice regarding implementation, particularly when writing the software in Rust.

Gaussian parameters should generally be stored in dense arrays (using linear algebra crates such as `nalgebra`). However, matrices within observation models are usually stored separately for each variable. Because these matrices typically become sparse as the graph grows, it is generally better to use sparse matrix collections tied to the variables they attach to.

From a memory management perspective – leveraging Rust's strict ownership model – messages constructed by summarizing multiple factors (priors and observation models) can simply *borrow* (e.g., via shared references `&T`) the immutable transformation matrices from the factors they summarize. However, the newly computed Gaussian parameters resulting from this summarization must be *owned* by the message itself to ensure memory safety and avoid lifetime entanglement.